Notes: A Parallel Chow Theory for Matroid Deletion

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Abstract

These notes develop an algebraic framework for studying the behavior of Chow polynomials of matroids under single-element deletion, in direct analogy with the Kazhdan-Lusztig theory of Braden and Vysogorets. We define a new basis $\{\mathrm{c}^F\}$ for the module of formal sums of flats, built from Chow polynomials. We then define a 'perverse-like' $\mathrm{Pal}(0)$-module $\mathcal{S}(\mathrm{M})$ for which this new basis is natural. We establish the main properties of this module and its basis, culminating in a conjectural deletion formula for $\Delta(\mathrm{c}^F)$. As an application, we show how this formula recovers the known semi-small decomposition formulas for both the regular and augmented Chow polynomials.

1 Definitions

Let M be a matroid of rank r with lattice of flats $\mathcal{L}(\mathrm{M})$. The module $\mathcal{H}(\mathrm{M})$ is defined as the free $\mathbb{Z}[x, x^{-1}]$-module with a standard basis $\{\mathrm{F}\}_{F\in\mathcal{L}(\mathrm{M})}$. Following Braden-Vysogorets [BV20], there is an important abelian subgroup $\mathcal{H}_p(\mathrm{M}) \subset \mathcal{H}(\mathrm{M})$ of **perverse elements**, consisting of all $\alpha = \sum \alpha_F \cdot \mathrm{F}$ such that for every flat $F \in \mathcal{L}(\mathrm{M})$:

1. The coefficient α_F is a polynomial in x, i.e., $\alpha_F \in \mathbb{Z}[x]$.
2. The Laurent polynomial $S_F^{\mathrm{M}}(\alpha)(x) := \sum_{G\geq F} x^{\mathrm{rk}(F)-\mathrm{rk}(G)}\alpha_G$ is in $\mathrm{Pal}(0)$.

Definition 1. *We relax the first condition to define a larger subgroup $\mathcal{H}'_p(\mathrm{M}) \subseteq \mathcal{H}(\mathrm{M})$ consisting of all elements α that satisfy the second (palindromicity) condition.*

Definition 2. *We define a new basis of $\mathcal{H}(\mathrm{M})$, analogous to the Kazhdan-Lusztig basis, given by*

$$\mathrm{c}^F = \sum_{G\leq F} \mathrm{c}^F_G \cdot \mathrm{G} := \sum_{G\leq F} x^{\mathrm{rk}(F)-\mathrm{rk}(G)} \underline{\mathrm{H}}_{\mathrm{M}^F_G}(x^{-2}) \cdot \mathrm{G}$$

where $\underline{\mathrm{H}}_{\mathrm{M}}$ is the Chow polynomial of the matroid M*, which is palindromic of degree* $\mathrm{rk}(\mathrm{M}) - 1$ *[Fer+24].*

Remark. The coefficients of this basis have a special structure. We use $\mathrm{Pal}(n)$ to denote the additive group of Laurent polynomials satisfying $f(x) = x^n f(x^{-1})$. A direct calculation shows:

$$\mathrm{c}^F_F = 1 \in \mathrm{Pal}(0), \quad \text{and} \quad \mathrm{c}^F_G \in \mathrm{Pal}(2) \text{ for all } G < F.$$

Definition 3. *Let $\mathcal{S}(\mathrm{M})$ be the free $\mathrm{Pal}(0)$-module with basis $\{\mathrm{c}^F\}$.*

Definition 4. *For an element e of* M*, the **deletion operator** is a $\mathbb{Z}[x, x^{-1}]$-module homomorphism $\Delta : \mathcal{H}(\mathrm{M}) \to \mathcal{H}(\mathrm{M} \setminus e)$ defined by*

$$\Delta(\mathrm{F}) = x^{-\delta(F)} \cdot \mathrm{F} \setminus \mathrm{e}, \quad \textit{where } \delta(F) := \mathrm{rk}_{\mathrm{M}}(F) - \mathrm{rk}_{\mathrm{M}\setminus e}(F \setminus e).$$

Definition 5. $S_e(\mathrm{M}) := \{F \in \mathcal{L}(\mathrm{M}) \mid e \notin F \text{ and } F \cup e \in \mathcal{L}(\mathrm{M})\}$.

2 Main Propositions

Proposition 6. *For all* $F \in \mathcal{L}(\mathrm{M})$, $\mathrm{c}^F \in \mathcal{H}'_p(\mathrm{M})$.

Proof. For any flat $G \leq F$, the palindromic check sum $S_G(\mathrm{c}^F)$ simplifies to $x^{\mathrm{rk}(F)-\mathrm{rk}(G)}\mathrm{H}_{\mathrm{M}^F_G}(x^{-2})$. Let $d = \mathrm{rk}(F) - \mathrm{rk}(G)$. The term x^d is in $\mathrm{Pal}(2d)$ and $\mathrm{H}_{\mathrm{M}^F_G}(x^{-2})$ is in $\mathrm{Pal}(-2d)$, so their product is in $\mathrm{Pal}(0)$. □

Proposition 7. *The subgroup* $\mathcal{H}'_p(\mathrm{M})$ *is a* $\mathrm{Pal}(0)$*-module.*

Proof. Let $\alpha \in \mathcal{H}'_p(\mathrm{M})$ and $q \in \mathrm{Pal}(0)$. The palindromic check sum for the product is $S_G(q \cdot \alpha) = q \cdot S_G(\alpha)$. Since $\mathrm{Pal}(0)$ is a ring, the product of two $\mathrm{Pal}(0)$ elements is in $\mathrm{Pal}(0)$. □

Proposition 8. $\mathcal{S}(\mathrm{M}) = \mathcal{H}'_p(\mathrm{M})$ *as* $\mathrm{Pal}(0)$*-modules.*

Proof. The inclusion $\mathcal{S}(\mathrm{M}) \subseteq \mathcal{H}'_p(\mathrm{M})$ follows from the previous two propositions. For the reverse, let $\alpha \in \mathcal{H}'_p(\mathrm{M})$. Since $\{\mathrm{c}^F\}$ is a triangular basis for $\mathcal{H}(\mathrm{M})$, α has a unique expansion $\alpha = \sum q_F \cdot \mathrm{c}^F$. An induction on decreasing rank shows that the condition $\alpha \in \mathcal{H}'_p(\mathrm{M})$ forces each coefficient q_F to be in $\mathrm{Pal}(0)$, which means $\alpha \in \mathcal{S}(\mathrm{M})$. □

Proposition 9. *Any Laurent polynomial* f *can be uniquely written as a sum* $f = f_0 + f_2$ *where* $f_0 \in \mathrm{Pal}(0)$ *and* $f_2 \in \mathrm{Pal}(2)$*. The projections are given by*

$$f_0(x) = \frac{x^2 f(x^{-1}) - f(x)}{x^2 - 1} \quad \textit{and} \quad f_2(x) = \frac{f(x) - f(x^{-1})}{1 - x^{-2}}.$$

Proposition 10 (Change of Basis Formula)**.** *If* $\alpha = \sum \alpha_F \cdot \mathrm{F} \in \mathcal{S}(\mathrm{M})$*, its* c^F*-basis expansion is* $\alpha = \sum q_F \cdot \mathrm{c}^F$*, where* q_F *is the unique* $\mathrm{Pal}(0)$ *component of* α_F *from the decomposition in Proposition 9.*

Proof. Since $\alpha \in \mathcal{S}(\mathrm{M})$, its c^F-basis coefficients q_F are in $\mathrm{Pal}(0)$. The change of basis formula gives $\alpha_F = q_F + \sum_{G>F} q_G \cdot \mathrm{c}^G_F$. The sum is an element of $\mathrm{Pal}(2)$. Thus, α_F is expressed as the sum of a $\mathrm{Pal}(0)$ element (q_F) and a $\mathrm{Pal}(2)$ element. By uniqueness (Proposition 9), q_F must be the $\mathrm{Pal}(0)$ part of α_F. □

Proposition 11. *For any flat* $F \in \mathcal{L}(\mathrm{M})$*, the element* $\Delta(\mathrm{c}^F)$ *is in* $\mathcal{S}(\mathrm{M} \setminus e)$*.*

Proof. This follows the proof of Proposition 2.14 in [BV20]. Since $\mathcal{S}(\mathrm{M} \setminus e) = \mathcal{H}'_p(\mathrm{M} \setminus e)$, we only need to check the palindromicity condition. The check sum $S_H(\Delta(\mathrm{c}^F))$ for any $H \in \mathcal{L}(\mathrm{M} \setminus e)$ simplifies, via the definition of discrepancy and Lemma 2.15 of [BV20], to the palindromicity check $S_{\overline{H}}(\mathrm{c}^F)$ for c^F at the flat $\overline{H} \in \mathcal{L}(\mathrm{M})$. Since $\mathrm{c}^F \in \mathcal{H}'_p(\mathrm{M})$, this is in $\mathrm{Pal}(0)$. □

3 The Deletion Formula and Its Consequences

The following proposition provides an explicit formula for the action of Δ on the c^F basis. It is the central result of this framework.

Proposition 12 (Deletion Formula)**.** *For any flat $F \in \mathcal{L}(\mathrm{M})$, the following identities hold in $\mathcal{S}(\mathrm{M} \setminus e)$:*

1. *If e is not a coloop in M^F, then*

$$\Delta(\mathrm{c}^F) = \mathrm{c}^{F\setminus e} + x^{-1} \sum_{G \in S_e(\mathrm{M}^F)} \mathrm{c}^F_{G\cup e} \cdot \mathrm{c}^G \tag{1}$$

2. *If e is a coloop in M^F, then*

$$\Delta(\mathrm{c}^F) = (x + x^{-1})\mathrm{c}^{F\setminus e} + x^{-1} \sum_{G \in S_e(\mathrm{M}^F), G \neq F\setminus e} \mathrm{c}^F_{G\cup e} \cdot \mathrm{c}^G \tag{2}$$

Proof. By Proposition 10, the coefficients q_G in the expansion $\Delta(\mathrm{c}^F) = \sum q_G \cdot \mathrm{c}^G$ are the unique $\mathrm{Pal}(0)$ components of the standard-basis coefficients β_G. A case analysis of β_G for $G \neq F \setminus e$ and $G = F \setminus e$ confirms the coefficients in the proposition. In particular, for $G = F \setminus e$, the standard-basis coefficient $\beta_{F\setminus e}$ is 1 in the non-coloop case and $\mathrm{c}^F_{F\setminus e} + x^{-1}$ in the coloop case. Extracting the $\mathrm{Pal}(0)$ parts of these expressions yields 1 and $x + x^{-1}$, respectively. □

3.1 Application 1: Recovery of the Semi-Small Decomposition

As a verification of the formula, we can recover the known semi-small decomposition formulas for $\underline{\mathrm{H}}$ [Fer+24] by taking the coefficient of the empty flat $\emptyset$ on both sides of (1) and (2). Let $y = x^{-2}$.

Decomposition Formulas for $\underline{\mathrm{H}}$

1. If e is not a coloop in M^F,

$$\underline{\mathrm{H}}_{\mathrm{M}^F}(y) = \underline{\mathrm{H}}_{\mathrm{M}^F\setminus e}(y) + y \sum_{G \in S_e(\mathrm{M}^F)\setminus\{\emptyset\}} \underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(y) \cdot \underline{\mathrm{H}}_{\mathrm{M}^G}(y).$$

2. If e is a coloop in M^F,

$$\underline{\mathrm{H}}_{\mathrm{M}^F}(y) = (1+y)\underline{\mathrm{H}}_{\mathrm{M}^F\setminus e}(y) + y \sum_{G \in \mathcal{L}(\mathrm{M}^F\setminus e)\setminus\{\emptyset\}} \underline{\mathrm{H}}_{(\mathrm{M}^F\setminus e)/G}(y) \cdot \underline{\mathrm{H}}_{(\mathrm{M}^F\setminus e)|_G}(y).$$

3.2 Application 2: New Formulas for Augmented Chow Polynomials

We consider the map $\Phi_{\mathrm{M}}(\alpha) = \sum_{F\in\mathcal{L}(\mathrm{M})} x^{-\mathrm{rk}(F)}\alpha_F$, which satisfies $\Phi_{\mathrm{M}\setminus e} \circ \Delta = \Phi_{\mathrm{M}}$. Applying this map to the deletion formulas in Proposition 12 yields new recursive identities for the augmented Chow polynomials.

Decomposition Formulas for H

1. If e is not a coloop in M^F,

$$\mathrm{H}_{\mathrm{M}^F}(y) = \mathrm{H}_{\mathrm{M}^F\setminus e}(y) + \sum_{G\in S_e(\mathrm{M}^F)} y\underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(y)\cdot \mathrm{H}_{\mathrm{M}^G}(y).$$

2. If e is a coloop in M^F,

$$\mathrm{H}_{\mathrm{M}^F}(y) = (1+y)\mathrm{H}_{\mathrm{M}^F\setminus e}(y) + \sum_{G\in S_e(\mathrm{M}^F)\setminus\{F\setminus e\}} y\underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(y)\cdot \mathrm{H}_{\mathrm{M}^G}(y).$$

4 Conclusion

This framework successfully establishes a parallel between the Kazhdan-Lusztig theory of matroids and a new theory based on Chow polynomials. The identification of the module $\mathcal{S}(\mathrm{M}) = \mathcal{H}'_p(\mathrm{M})$ and its natural basis $\{\mathrm{c}^F\}$ provides a solid foundation. The main result is a conjectural deletion formula that behaves predictably on this module, which in turn implies new and known identities among Chow and augmented Chow polynomials. The primary remaining task is a direct, combinatorial proof of the standard-basis coefficient identities that underpin the main deletion formula.

References

[BV20] Tom Braden and Artem Vysogorets. "Kazhdan-Lusztig polynomials of matroids under deletion". In: *Electron. J. Combin.* 27.1 (2020), Paper No. 1.17.

[Fer+24] Luis Ferroni et al. "Hilbert-Poincaré series of matroid Chow rings and intersection cohomology". In: (2024). arXiv: 2212.03190v3 [math.CO].